Lincoln 80th Lights (2025)

Joshua Oliveira, *student, Loyola Marymount University*

***Abstract*—This experiment implemented a traffic light controller for the intersection of Loyola Boulevard and 80th Street using a finite state machine (FSM) on a ZedBoard FPGA. The design prioritized traffic on Loyola Blvd due to higher vehicle flow, with adaptive timing based on car sensor inputs. A 0.5 Hz clock signal was derived through frequency division of the ZedBoard's 100 MHz input clock, requiring a division factor of 200,000,000. A 14-state FSM was developed to manage light transitions effectively, accounting for extended green periods when vehicles were detected. The design was simulated using a testbench with an adjusted clock frequency to accelerate evaluation. Implementation and real-time testing confirmed correct FSM behavior and verified timing functionality, showcasing FPGA advantages in prototyping and real-world traffic control applications.**

I. INTRODUCTION

Traffic management systems play a vital role in reducing congestion, minimizing delays, and improving road safety. The objective of this experiment was to design and implement a traffic light controller for a two-way intersection at Loyola Boulevard and 80th Street, prioritizing the busier Loyola Blvd while adapting to real-time vehicle presence using sensor inputs. The controller was implemented on a ZedBoard FPGA using VHDL, enabling precise timing control and rapid state transitions via a finite state machine (FSM).

A total of 14 states were required to handle all traffic light transitions, pedestrian intervals, and sensor-based extensions. To achieve the desired 2-second clock period for FSM transitions, the ZedBoard's 100 MHz clock was divided using a programmable clock divider with a factor of 200,000,000. This setup ensured that light phases adhered to specified durations: Loyola Blvd green for 14 seconds (extendable by 4 seconds), 80th Street green for 10 seconds, and yellow transitions for 2 seconds.

This experiment provided an opportunity to explore digital system design with FPGAs and examine their benefits, such as design flexibility and real-time testing capabilities. Furthermore, traffic light timing decisions were considered within economic, environmental, and societal contexts. Efficient traffic control has been shown to reduce vehicle idle times, lower carbon emissions, and mitigate accident risks—key metrics that underscore the real-world significance of this work.

II. SYSTEM DESCRIPTION

The implemented system is a traffic light controller designed using a finite state machine (FSM) architecture programmed on a ZedBoard FPGA. The controller manages traffic signals at the intersection of Loyola Boulevard and 80th Street, with Loyola Blvd receiving priority due to higher traffic volume. The system responds dynamically to vehicle presence using two input sensors, one for each direction.

A. Clock Division

The ZedBoard's 100 MHz base clock was divided to generate a 0.5 Hz signal, corresponding to a 2-second state duration. A clock divider module was implemented in VHDL, using a FACTOR value of 200,000,000 to achieve the required frequency. This divided clock signal was used to synchronize all FSM transitions.

B. FSM Structure

The FSM was modeled as a Moore machine with 14 states, each representing a unique light configuration and timing condition. The states were designed to:

- Maintain green light for Loyola Blvd for 14 seconds, extendable by 4 seconds if vehicles are detected.
- Provide a 10-second green light for 80th Street.
- Include 2-second yellow light phases for both streets before transitions.
- Handle all transitions in compliance with sensor inputs and preset durations.

State transitions were determined by current state, timing constraints, and real-time sensor inputs. FSM states were encoded using one-hot encoding to simplify logic and reduce transition errors.

C. Signal Outputs

The outputs of the FSM consisted of two 3-bit vectors (loyola_tl and st80th_tl) representing the traffic light colors for each direction. Each vector followed the standard color encoding: 001 for green, 010 for yellow, and 100 for red. These outputs were mapped to the physical traffic-light board connected to the ZedBoard via PMOD interfaces.

D. Simulation

Simulation of the FSM was performed using a VHDL testbench. To expedite simulation, the clock division factor was set to 1, allowing state transitions to occur at the base clock frequency. The simulation verified all transitions and light sequences according to the FSM design, confirming correct logic implementation and responsiveness to sensor input.

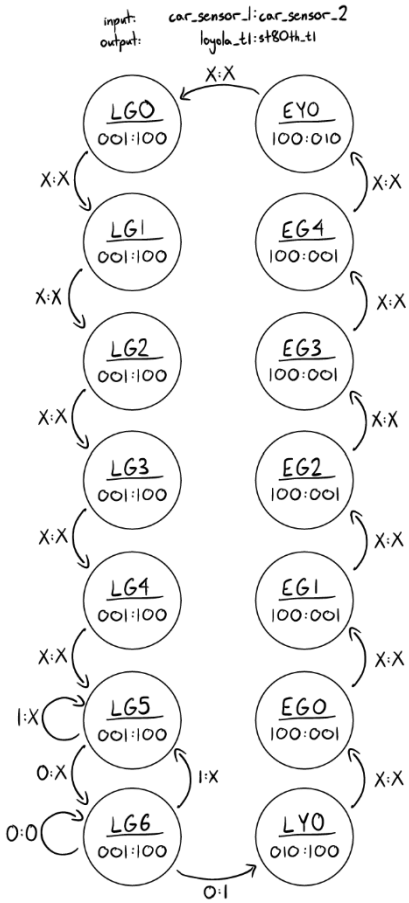


Figure 1: Moore Model Finite State Machine Diagram

IV. Experimental Evaluation

A. Performance Metrics

The traffic light controller was evaluated using the following performance metrics:

- **Clock Accuracy**: Ability to maintain a 2-second cycle using a 0.5 Hz signal derived from the 100 MHz base clock.
- **State Coverage**: Validation that all 14 FSM states were correctly reached and exited.
- **Output Accuracy**: Correct generation of light signals (loyola_tl, st80th_tl) according to the FSM logic and timing.
- **Sensor Responsiveness**: Proper extension of the green phase on Loyola Blvd based on active vehicle detection.

B. List of Materials

- ZedBoard FPGA development board
- ZedBoard power adapter
- USB–MicroUSB cable
- Traffic-light board
- PMOD cables (2)
- Vivado Design Suite (Xilinx)

C. Experimental Procedure

The experimental process began with the design of a finite state machine (FSM) in VHDL to manage the traffic light sequence based on timing constraints and sensor inputs. A 14-state Moore machine was implemented to account for all required transitions, including standard green, yellow, and red intervals, as well as conditional green extensions based on vehicle detection.

To maintain a 2-second cycle time per FSM state, a clock divider module was developed. This module used a counter to divide the 100 MHz input clock from the ZedBoard, with a division factor of 200,000,000, producing a 0.5 Hz output signal. The FSM operated on this divided clock to ensure timing precision across all light phases.

The VHDL code was simulated using Vivado's integrated simulation tools. A dedicated testbench was written to verify functional behavior of the FSM under accelerated conditions by setting the division factor to 1. This allowed state transitions to occur quickly, enabling rapid validation of all transitions and output logic. During simulation, waveform outputs were analyzed to confirm correct sequencing, light color outputs, and state transitions in response to simulated sensor inputs.

Following successful simulation, the VHDL design was synthesized and implemented in Vivado. Output ports for the traffic light signals (loyola_tl and st80th_tl) and input ports for the clock and sensor signals were assigned to the appropriate physical pins on the ZedBoard using Vivado's constraints editor. These mappings enabled correct electrical connections between the FPGA and the external traffic light module via PMOD interfaces.

After synthesis and implementation, the bitstream was generated and programmed onto the ZedBoard. The physical hardware was then connected using two PMOD cables to interface the FSM outputs with the traffic light board. System functionality was validated in real-time, with the state transitions, timing behavior, and sensor-driven logic observed directly through the behavior of the traffic lights on the board.

D. Results

The traffic light controller operated correctly in both simulation and hardware testing. The clock divider accurately produced a 0.5 Hz signal from the ZedBoard's 100 MHz clock, enabling 2-second state intervals. All 14 FSM states were verified through simulation and were traversed in the correct sequence, with proper timing and output transitions.

Simulation using Vivado confirmed that the FSM responded appropriately to sensor inputs and that light signals (loyola_tl, st80th_tl) transitioned as designed. To prevent indefinite green time for Loyola Blvd when cars were present on both streets, the VHDL code was modified. The extension of the green phase for Loyola was limited to a single 4-second period; afterward, control shifted to 80th Street, regardless of continued car presence, ensuring fair access.

In hardware testing, PMOD connections successfully mapped FSM outputs to the traffic light board. Real-time observations confirmed that transitions and timing matched simulation results. The system responded accurately to sensor inputs, and the modified logic prevented starvation of 80th Street under persistent traffic conditions.

E. Discussion of Results

The traffic light controller met all functional and timing requirements in both simulation and hardware testing. The FSM transitioned correctly through all 14 states, and the 0.5 Hz clock division ensured accurate 2-second intervals. Simulation using a reduced clock factor accelerated development and helped verify state behavior efficiently.

One issue encountered during testing was that Loyola Blvd's green light extended indefinitely when cars were present on both streets. To resolve this, the VHDL code was modified to allow only a single 4-second extension, after which the light switched to 80th Street. This change ensured fair traffic handling and improved system reliability. All other aspects of the design performed as expected.

F. Limitations

The primary limitation was the slow testing cycle caused by the 2-second state duration. Although simulation with FACTOR = 1 mitigated this during development, physical testing required patience to observe transitions. Additionally, the system is tailored to specific timing constraints and lacks configurability for different intersections without reprogramming.

V. Conclusion

The design and implementation of the traffic light controller successfully demonstrated the use of finite state machines (FSMs) on an FPGA platform for real-time digital control systems. The final system met all timing and functional specifications, with accurate 2-second state intervals and correct transitions based on sensor inputs. Simulation and hardware results confirmed that the FSM logic was sound and the Vivado implementation was effective.

A key improvement was the modification of the green light extension logic for Loyola Blvd. Initially, persistent vehicle detection led to indefinite green phases, preventing access for 80th Street. Updating the code to limit the extension to one cycle improved fairness and system responsiveness. This highlights the importance of iterative testing and refinement during the design process

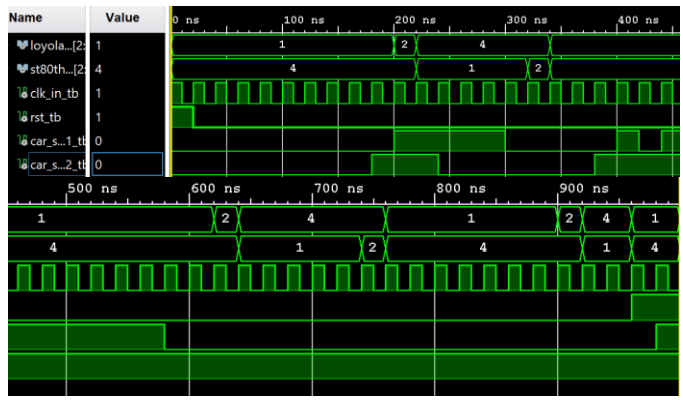
Overall, the experiment reinforced key concepts in digital design, including clock management, FSM construction, and hardware-software integration. The lab also illustrated the practical advantages of FPGA-based systems, particularly in applications where real-time responsiveness and reconfigurability are critical, such as traffic control.

References

[1] N. Ula, Introduction to ARM Assembly Language programming using Raspberry Pi, Loyola Marymount University, EECE3200-Lab01Tutorial (1).pdf, 2025

[2] OpenAI, ChatGPT, Loyola Marymount University, ChatGPT, 2025

ATTACHMENTS



Simulation

On Brightspace Vivado Full Project with Files is Attached